Naifeng Zhang

naifengz@cmu.edu
naifengz.com

EDUCATION

2021–27 **Carnegie Mellon University**
Ph.D. in Electrical and Computer Engineering
Advisor: Franz Franchetti

2021–24 **Carnegie Mellon University**
M.S. in Electrical and Computer Engineering
Advisor: Franz Franchetti

2017–21 **University of Southern California**
B.S. in Computer Science
Advisor: Viktor K. Prasanna
Thesis: Lightweight Augmented Neural Network For Performance Prediction and Its Applications
W.V.T. Rusch Undergraduate Engineering Honors Program

2017–21 **University of Southern California**
B.S. in Mathematics
Departmental Honors Program

EXPERIENCE

2021- **Carnegie Mellon University**, SPIRAL Group
Graduate Researcher, Pittsburgh, PA

- Developing code generation systems targeting CPUs [C4, W3], GPUs [M1, C6, W8], and specialized hardware [C7, C9] for high-performance scientific and cryptographic computing.
- Developing semantics lifting techniques for program reasoning, enabling code modernization and verification of AI-generated scientific code [W1, W2].

2026 **NVIDIA**, CUDA Math Libraries
Intern, Remote

- Prototyping a code-generation framework for CUDA Tile implementations of high-performance math library kernels.

2025 **NVIDIA**, Programming Systems and Applications Research Group
Intern, Santa Clara, CA

- Extended [C6] with architecture-aware optimizations for NVIDIA cryptographic workloads.
- Achieved near-peak performance across multiple NVIDIA GPUs [M1]; selected for Chief Scientist Bill Dally's company-wide "Top-5 Things."

2018–21 **University of Southern California**, Data Science Lab
Undergraduate Researcher, Los Angeles, CA

- Developed lightweight neural-network-based performance prediction methods [C11] and a prediction-guided auto-tuner [C10] for heterogeneous platforms.

GRANTS

2023–26 **NSF ACCESS**, *High-Performance Code Generation for Homomorphic Encryption on GPUs using SPIRAL*
N. Zhang (PI), F. Franchetti (Co-PI)

- Awarded 200,000 ACCESS Credits for tuning, benchmarking, and extending SPIRAL-generated cryptographic kernels for homomorphic encryption on state-of-the-art CPUs and GPUs.

PUBLICATIONS

Underlined names denote undergraduate and Master's mentees.

Conference Proceedings

1. S. Rao, A. Prakash, **N. Zhang**, H. Mankad, F. Franchetti. "LibraryX: A Framework for Cross-Library-Call Optimization." The IEEE International Parallel & Distributed Processing Symposium (**IPDPS**), 2026.
2. **N. Zhang**, S. McAleer, T. Sandholm. "Faster Game Solving via Hyperparameter Schedules." The AAAI Conference on Artificial Intelligence (**AAAI**), 2026.
3. Y. Lan, L. Tang, **N. Zhang**, Y. Eum, J. Hoe, F. Franchetti. "A RISC-V Vector Extension for Multi-word Arithmetic." The International Workshop on RISC-V for HPC (**RISCV-HPC**), in conjunction with the International Conference for High Performance Computing, Networking, Storage, and Analysis (**SC**), 2025.
4. **N. Zhang**, S. Fu, F. Franchetti. "Towards Closing the Performance Gap for Cryptographic Kernels Between CPUs and Specialized Hardware" The IEEE/ACM International Symposium on Microarchitecture (**MICRO**), 2025.
5. Q. Oschatz, **N. Zhang**, M. Franusich, F. Franchetti. "Towards Automated Reasoning Chains for Verification of LLM-Generated Scientific Code." IEEE High Performance Extreme Computing Conference (**HPEC**), 2025.
6. **N. Zhang**, F. Franchetti. "Code Generation for Cryptographic Kernels using Multi-word Modular Arithmetic on GPU." The International Symposium on Code Generation and Optimization (**CGO**), 2025.
7. **N. Zhang**, A. Ebel, N. Neda, P. Brinich, B. Reynwar, A. G. Schmidt, M. Franusich, J. Johnson, B. Reagen, F. Franchetti. "Generating High-Performance Number Theoretic Transform Implementations for Vector Architectures." IEEE High Performance Extreme Computing Conference (**HPEC**), 2023.
8. D. Sun, **N. Zhang**, F. Franchetti. "Optimization and Performance Analysis of Shor's Algorithm in Qiskit." IEEE High Performance Extreme Computing Conference (**HPEC**), 2023.
9. D. Soni, N. Neda, **N. Zhang**, B. Reynwar, H. Gamil, B. Heyman, M. N. T. Moopan, A. Al Badawi, Y. Polyakov, K. Canida, M. Pedram, M. Maniatakos, D. B. Cousins, F. Franchetti, M. French, A. Schmidt, B. Reagen. "RPU: The Ring Processing Unit." IEEE International Symposium on Performance Analysis of Systems and Software (**ISPASS**), 2023.
10. **N. Zhang**, A. Srivastava, R. Kannan, V. K. Prasanna. "GenMAT: A General-Purpose Machine Learning-Driven Auto-Tuner for Heterogeneous Platforms." The Workshop on Programming Environments for Heterogeneous Computing (**PEHC**), in conjunction with the International Conference for High Performance Computing, Networking, Storage, and Analysis (**SC**), 2021.
11. A. Srivastava*, **N. Zhang***, R. Kannan, V. K. Prasanna. "Towards High Performance, Portability, and Productivity: Lightweight Augmented Neural Networks for Performance Prediction." The International Conference on High Performance Computing, Data, and Analytics (**HiPC**), 2020. **Equal contribution*.
12. C. Imes, A. Colin, **N. Zhang**, A. Srivastava, V. K. Prasanna, J. P. Walters. "Compiler Abstractions and Runtime for Extreme-scale SAR and CFD Workloads." The Workshop on Extreme Scale Programming Models and Middleware (**ESPM2**), in conjunction with the International Conference for High Performance Computing, Networking, Storage, and Analysis (**SC**), 2020.

Other Conference Papers, Extended Abstracts, and Posters

1. **N. Zhang**, F. Franchetti. "Semantics Lifting for Scientific Kernels." The ACM SIGPLAN Conference on Programming Language Design and Implementation (**PLDI**), 2026, Extended abstract with poster and presentation. **Third Place, ACM Student Research Competition**.

2. **N. Zhang**, S. Rao, M. Franusich, F. Franchetti. "Towards Semantics Lifting for Scientific Computing: A Case Study on FFT." The Theory and Practice of Static Analysis Workshop (**TPSA**), in conjunction with the ACM SIGPLAN Symposium on Principles of Programming Languages (**POPL**), 2025, Extended abstract with presentation.

3. S. Fu, **N. Zhang**, F. Franchetti. "Accelerating High-Precision Number Theoretic Transforms using Intel AVX-512." The International Conference on Parallel Architectures and Compilation Techniques (**PACT**), 2024, Extended abstract with poster and presentation. **First Place, ACM Student Research Competition**. **Best Poster Runner-up** at PRISM Annual Review, Systems & Software track.

4. Y. Eum, **N. Zhang**, L. Tang, F. Franchetti. "Towards a RISC-V Instruction Set Extension for Multi-word Arithmetic." IEEE High Performance Extreme Computing Conference (**HPEC**), 2024, Extended abstract with poster.

5. P. Brinich, **N. Zhang**, A. Ebel, F. Franchetti, J. Johnson. "Twiddle Factor Generation for a Vectorized Number Theoretic Transform." IEEE High Performance Extreme Computing Conference (**HPEC**), 2023, Extended abstract with poster. **Outstanding Short Paper Award**.

6. H. Mankad, A. Rovinelli, M. Zecevic, P. McCorquodale, F. Franchetti, **N. Zhang**, S. Rao, R. A. Lebensohn, L. Capolungo. "EVPFFTX: A First Look at FFTX Applications in Material Science." IEEE High Performance Extreme Computing Conference (**HPEC**), 2023, Extended abstract with poster.

7. D. B. Cousins, Y. Polyakov, A. Al Badawi, M. French, A. Schmidt, A. Jacob, B. Reynwar, K. Canida, A. Jaiswal, C. Mathew, H. Gamil, N. Neda, D. Soni, M. Maniatakos, B. Reagen, **N. Zhang**, F. Franchetti, P. Brinich, J. Johnson, P. Broderick, M. Franusich, B. Zhang, Z. Cheng, M. Pedram. "TREBUCHET: Fully Homomorphic Encryption Accelerator for Deep Computation." The Government Microcircuit Applications and Critical Technology Conference (**GOMACTech**), 2023, Preprint with presentation.

8. **N. Zhang**, F. Franchetti. "Generating Number Theoretic Transforms for Multi-Word Integer Data Types." The International Symposium on Code Generation and Optimization (**CGO**), 2023, Extended abstract with poster and presentation. **Second Place, ACM Student Research Competition**.

9. **N. Zhang**, H. Gamil, P. Brinich, B. Reynwar, A. Al Badawi, N. Neda, D. Soni, K. Canida, Y. Polyakov, P. Broderick, M. Maniatakos, A. G. Schmidt, M. Franusich, J. Johnson, B. Reagen, D. B. Cousins, F. Franchetti. "Towards Full-Stack Acceleration for Fully Homomorphic Encryption." IEEE High Performance Extreme Computing Conference (**HPEC**), 2022, Extended abstract with presentation.

10. I. Grosof, **N. Zhang**, M. Heule. "Towards the shortest DRAT proof of the Pigeonhole Principle." The Pragmatics of SAT Workshop (**PoS**), in conjunction with the International Conference on Theory and Applications of Satisfiability Testing (**SAT**), 2022, Preprint with presentation.

Manuscripts Under Review

1. **N. Zhang**, C. Gouert, S. Dalton, V. Joseph, M. Garland, F. Franchetti. "Architecture-Aware Code Generation for Number Theoretic Transforms on GPUs."

2. S. Alsultan, P. Brinich, **N. Zhang**, J. Johnson. "Performance Analysis of Vectorized Number Theoretic Transforms."

TALKS

Seminars

2026 *Code Generation for Cryptographic Kernels using Multi-word Modular Arithmetic*
Carnegie Mellon University, Graduate Student and Postdoc Seminars (GSPS), Department of Mathematical Sciences, Feb. 26

2026 *Closing the Performance Gap for Cryptographic Kernels via Code Generation*
University of California, Los Angeles, VLSI Architecture, Synthesis, and Technology (VAST) Laboratory, Jan. 27

2025 *Towards Closing the Performance Gap for Cryptographic Kernels Between CPUs and Specialized Hardware*
Carnegie Mellon University, Computer Architecture Lab at Carnegie Mellon (CALCM), Oct. 2

2025 *Code Generation for Cryptographic Kernels using Multi-word Modular Arithmetic*
University of Southern California, Ming Hsieh Department of Electrical and Computer Engineering, May 9

2025 *Code Generation for Cryptographic Kernels using Multi-word Modular Arithmetic*
New York University, Department of Electrical and Computer Engineering, May 2

2025 *Optimization and Performance Analysis of Shor's Algorithm in Qiskit and Beyond*
Carnegie Mellon University, The Center for Quantum Computing and Information Technologies (QCiT), Apr. 1

2025 *Code Generation for Cryptographic Kernels using Multi-word Modular Arithmetic*
University of Pennsylvania, The Programming Languages Group at the University of Pennsylvania (PLClub), Feb. 21

2025 *Code Generation for Cryptographic Kernels using Multi-word Modular Arithmetic*
Carnegie Mellon University, Computer Architecture Lab at Carnegie Mellon (CALCM), Feb. 14

Guest Lectures

2025 *Code Generation for Cryptographic Kernels using Multi-word Modular Arithmetic*
Carnegie Mellon University, Computational Problem Solving for Engineers, Apr. 1

Conference Presentations

2026 *Semantics Lifting for Scientific Kernels*
The ACM SIGPLAN Conference on Programming Language Design and Implementation (**PLDI**), Boulder, CO, Jun. 18

2026 *Semantics Lifting for Quantum Circuits*
Penn Forum on Quantum Systems (**Penn FoQuS**), Philadelphia, PA, Apr. 21

2025 *Towards Closing the Performance Gap for Cryptographic Kernels Between CPUs and Specialized Hardware*
The IEEE/ACM International Symposium on Microarchitecture (**MICRO**), Virtual, Oct. 22

2025 *Code Generation for Cryptographic Kernels using Multi-word Modular Arithmetic*
The Workshop on Architectures for Zero-Knowledge Proofs and Verifiable Computation (**ZKARCH**), in conjunction with the IEEE/ACM International Symposium on Microarchitecture (**MICRO**), Virtual, Oct. 18

2025 *Towards Semantics Lifting for Scientific Computing: A Case Study on FFT*
ORNL AI4Science Workshop, Oak Ridge, TN, Apr. 30

2025 *Code Generation for Cryptographic Kernels using Multi-word Modular Arithmetic on GPU*
The International Symposium on Code Generation and Optimization (**CGO**), Las Vegas, NV, Mar. 4

2025 *Towards Semantics Lifting for Scientific Computing: A Case Study on FFT*
The Theory and Practice of Static Analysis Workshop (**TPSA**), in conjunction with the ACM SIGPLAN Symposium on Principles of Programming Languages (**POPL**), Denver, CO, Jan. 21

2023 *Generating High-Performance Number Theoretic Transform Implementations for Vector Architectures*
IEEE High Performance Extreme Computing Conference (**HPEC**), Virtual, Sep. 29

2023 *Generating Number Theoretic Transforms for Multi-Word Integer Data Types*
The International Symposium on Code Generation and Optimization (**CGO**), Montreal, Canada, Feb. 28

2022 *Towards Full-Stack Acceleration for Fully Homomorphic Encryption*
IEEE High Performance Extreme Computing Conference (**HPEC**), Virtual, Sep. 23

2021 *GenMAT: A General-Purpose Machine Learning-Driven Auto-Tuner for Heterogeneous Platforms*
The Workshop on Programming Environments for Heterogeneous Computing (**PEHC**), in conjunction with the International Conference for High Performance Computing, Networking, Storage, and Analysis (**SC**), Virtual, Nov. 19

2020 *Towards High Performance, Portability, and Productivity: Lightweight Augmented Neural Networks for Performance Prediction*
The International Conference on High Performance Computing, Data, and Analytics (**HiPC**), Virtual, Dec. 16

AWARDS

2026 **Third Place, ACM Student Research Competition**, PLDI
For "Semantics Lifting for Scientific Kernels"

2024 **Best Poster Runner-up**, PRISM Annual Review, Systems & Software Track
For "Accelerating High-Precision Number Theoretic Transforms using Intel AVX-512"

2023 **Outstanding Short Paper Award**, HPEC
For "Twiddle Factor Generation for a Vectorized Number Theoretic Transform"

2023 **Second Place, ACM Student Research Competition**, CGO
For "Generating Number Theoretic Transforms for Multi-Word Integer Data Types"

2021 **Carnegie Institute of Technology Dean's Fellow**, Carnegie Mellon University

2021 **Discovery Scholar Distinction**, University of Southern California

2019–21 **Provost's Research Fellowship**, University of Southern California

2018–21 **Academic Achievement Award**, University of Southern California

TEACHING

Carnegie Mellon University

Teaching assistant

2024 Mathematical Foundations of Electrical Engineering

2023 Computational Problem Solving for Engineers

University of Southern California

Undergraduate teaching assistant

2021	Special Topics: Accelerated Computing Using FPGAs
2020	Parallel and Distributed Computation
2020	Special Topics: Accelerated Computing Using FPGAs
2020	Discrete Methods in Computer Science
2019	Parallel and Distributed Computation
2019	Discrete Methods in Computer Science

Conference Tutorials

Co-organizer and presenter

2026 — *SPIRAL Tutorial: A Code Generation Approach to Hardware-Software Co-Design*
ACM International Conference on Architectural Support for Programming Languages and Operating Systems (**ASPLOS**), Pittsburgh, PA, Mar. 23

2025 — *Open Source SPIRAL 8.5.1 Tutorial*
IEEE High Performance Extreme Computing Conference (**HPEC**), Virtual, Sep. 17

2024 — *Open Source SPIRAL 8.5 Tutorial*
IEEE High Performance Extreme Computing Conference (**HPEC**), Virtual, Sep. 25

2023 — *Open Source SPIRAL 8.5 Tutorial*
IEEE High Performance Extreme Computing Conference (**HPEC**), Virtual, Sep. 27

MENTORING

All mentees are from Carnegie Mellon University unless otherwise noted. Parentheses denote first known post-graduation affiliation.

Undergraduate

2026-	Jamie Kim
2025	Yiwen Jiang
2024–26	Sophia Fu (Apple)
2024–25	Misho Alexandrov
2024	Zubin Narayan
2024	Youngjin Eum
2024	Govind Malasani
2024	Steven Lee
2023–25	Gordon Xu (Etched)
2022–23	Matt Ngaw (Arm)
2022–23	Jimmy Zhou (NVIDIA)

Master's

2025–26	Yunhao Lan (CMU Ph.D.)
2024–25	Yujun Lee (Meta)
2023	Kofi Poku (PNNL)
2022–23	Dewang Sun (Cadence)
2022	Hongbo Sun (Alibaba)

Ph.D.

2026-	Matei Budiu, UIUC

SERVICE

Program Committees

The AAAI Conference on Artificial Intelligence (**AAAI**), 2026–27

The Workshop on Systems and Architectures for Encrypted AI (**SAFE-AI**), in conjunction with the International Symposium on Computer Architecture (**ISCA**), 2026

The Workshop on AI Assisted Software Development for HPC (**AI4Dev**), in conjunction with the International Conference on Parallel Processing (**ICPP**), 2025

Journal Peer Review

IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (**TCAD**), 2025–26

IEEE Transactions on Dependable and Secure Computing (**TDSC**), 2025–26

IEEE Transactions on Parallel and Distributed Systems (**TPDS**), 2025–26

IEEE Transactions on Mobile Computing (**TMC**), 2025–26

IEEE Transactions on Computational Social Systems (**TCSS**), 2025

IEEE Transactions on Emerging Topics in Computing (**TETC**), 2025

ACM Computing Surveys (**CSUR**), 2025

The International Journal of High Performance Computing Applications (**IJHPCA**), 2025

IEEE Transactions on Information Forensics & Security (**T-IFS**), 2025

Service to the University

Graduate Student Representative, College of Engineering Council, Carnegie Mellon University, 2025–26

Member, Department of Electrical and Computer Engineering Faculty Hiring Student Council, Carnegie Mellon University, 2022–25

Outreach

Member, College of Engineering Graduate Student Outreach Committee, Carnegie Mellon University, 2023

MEDIA

Naifeng Zhang: Researching High-Speed Computational Kernels, College of Engineering, Carnegie Mellon University, 2026

Updated June 2026